

Phaneendra Kumar Papinedi

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SUMMARY

Computer Science (IoT) graduate who has built end-to-end, real-time data systems: Python/OpenCV and YOLOv8 computer-vision pipelines, Raspberry Pi and ESP32 sensor platforms, and Pandas/NumPy analytics. Research Intern at IIIT Hyderabad and co-author of a DOI-indexed IoT paper. Looking to apply Python, SQL, data structures and AI/IoT engineering to large-scale client solutions at TCS.

TECHNICAL SKILLS

- **Languages:** Python, C, Java, SQL
- **Core CS (coursework):** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Computer Vision and Data:** OpenCV, YOLOv8 (object detection), image processing, NumPy, Pandas, time-series data handling
- **IoT and Embedded:** Raspberry Pi, ESP32/ESP8266, Arduino, sensor integration, data acquisition

EXPERIENCE

Research Intern, SPCRC, IIIT Hyderabad | Feb 2026 - Jul 2026

- Built an automated IoT pan evaporimeter for continuous, real-time monitoring of evaporation and environmental conditions.
- Developed a Python data-acquisition pipeline on Raspberry Pi that synchronises multiple environmental sensors for meteorological and evaporation measurements.
- Implemented periodic high-resolution image capture and Python/OpenCV pipelines to measure water-level changes in the evaporation pan from images.
- Cleaned, merged and analysed sensor and image-derived time-series data with Pandas and NumPy to identify evaporation patterns.

PROJECTS

Monkey Detection System | Python, YOLOv8, OpenCV

- Fine-tuned a pretrained YOLOv8n object detector for 50 epochs on a custom monkey dataset, reaching 0.82 mAP@0.5 and 0.54 mAP@0.5:0.95 on the validation split.
- Built a real-time inference script with OpenCV that runs the model on a webcam or video stream and overlays the detections with a live count.

Computer Vision Water Level Monitor | Python, OpenCV

- Automated lake water-level monitoring with an OpenCV pipeline, replacing manual observation with repeatable image analysis.
- Used region-of-interest segmentation and pixel-level measurement on reference markers to convert image data into water-level readings.

PUBLICATION

Hand Gesture Recognition for Specially Impaired People Using IoT, IJRASET, Vol. 14, Issue IV, April 2026, pp. 3009-3016 (DOI: [10.22214/ijraset.2026.80001](https://doi.org/10.22214/ijraset.2026.80001)). Co-authors: Kancherla Balakrishna, P. Guru Manikanta, P. Sai, P. Kavitha.

- Co-authored a paper on a wearable, threshold-based (no machine learning) gesture recognition system using an ESP32, five flex sensors and an MPU6050.
- The paper reports 96% recognition accuracy and 78 ms average response time in prototype testing.

EDUCATION

- **B.Tech, Computer Science and Engineering (Internet of Things)**, RVR & JC College of Engineering, Guntur | 2022 - 2026 | CGPA 6.89/10
- **Intermediate (MPC)**, Bhashyam Junior College | 2020 - 2022 | 71.1%
- **Secondary Education**, Loyola Public School | 2019 - 2020 | 71.0%